

HAKAN DURAN

Full Stack Software Engineer – Computer Engineer & Scientist –
Agentic AI & Cloud Infrastructure
See hakanduran.me for further details.

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EDUCATION

Technical University of Munich

Master of Science in Informatics

2022 – October 2026 (expected)

Munich, Germany

- Thesis: *An Agentic LLM System for Executive Decision Support over Hierarchical KPI Outlier Data*, carried out at Infineon.

Atilim University

Bachelor of Science in Computer Engineering, GPA: 3.68/4.00 (Top of the Department)

2016 – 2021

Ankara, Turkey

Poznan University of Technology

Exchange Student (Erasmus+) in Computer Science

2019 – 2020

Poznan, Poland

RECENT COURSEWORK

- Advanced Topics of SW Engineering
- Requirements Engineering
- Fundamentals of AI
- Computational Surgery
- Resilient Cognitive Systems
- DevOps Engineering
- Protein Prediction
- AWS Cloud Practitioner (Ongoing)

WORK EXPERIENCE

Infineon Technologies AG

AI Agent Intern & Master's Thesis Student · Full-time

March 2026 – September 2026

Munich, Germany

- Agentic LLM system for executive decision support (Python, LangGraph, LangChain, DuckDB, Impala/SQL):** a semantic-context builder, a weekly executive-summary pipeline, and a conversational agent that share one expert-approved semantic layer, one read-only data layer, and one memory store. Runs on **Claude Sonnet 5, Claude Opus 5, and GPT-5.6** through an OpenAI-compatible model gateway with structured outputs.
- Semantic-context builder:** constructs the semantic layer from schema, stored values, and company documentation through ingest, verified time semantics, and an LLM-driven challenge loop that turns failed questions into reviewable rules; **human-in-the-loop** approval, a searchable value lookup, and versioned rebuilds.
- Executive-summary pipeline:** turns the many flagged KPI outliers of a week into a short executive report. Bounded agents choose the drill-down level and the story selection inside tool-enforced guards; deterministic tools handle same-effect clustering on raw records, impact scoring, record ownership, and accounting, so every reported amount reconciles to the measured total. Published as JSON and PDF by a scheduled weekly job.
- Conversational agent:** a tool-first **orchestrator** that answers business-language questions over hierarchical KPI data, escalating from 20 deterministic analytics tools (selected by **progressive disclosure**, exposed via **MCP**) to guarded, read-only **Text-to-SQL** in a bounded **ReAct** loop; typed outcomes (answer, clarification, refusal), value grounding for business names, and conversation memory.
- Evaluation:** built the evaluation harness and ran it on company data, checking answers against independent reference queries and comparing the tool-first agent with direct Text-to-SQL on correctness and model usage; the tool-first design delivered more fully correct answers at lower cost, with every run traced and reproducible.
- Delivery:** a Dash web application (conversation, weekly reports, operator controls), MCP servers, Docker Compose deployment, a manifest-checked transfer to the company environment, and a Windows/Linux-portable evaluation launcher; requirements to implementation in design reviews with business and university stakeholders.

Everstox GmbH

IT & Automation Working Student · Part-time

June 2024 – February 2026

Munich, Germany

- Designed and implemented AI-powered automation solutions using **Large Language Model APIs** (ChatGPT, Perplexity) and workflow automation platforms (**Make.com, n8n, Airbyte**) to support cross-department workflows, enabling automated triage, data enrichment, and runbook execution.
- Provisioned and managed cloud infrastructure with **Terraform** and **Terraform Cloud**, including S3 buckets, Lambda functions, and secure credential management via **Secrets Manager + KMS** (least-privilege IAM).
- Implemented partner data pipelines S3 → Airbyte → BigQuery with S3 lifecycle policies, SSE-KMS encryption, and monitoring/alerting for reliability.

Technical University of Munich

Full Stack Web Developer – HiWi · Part-time

November 2022 – March 2024

Munich, Germany

- Participated in TUM's CommonRoad project, focusing on web application development and maintenance.
- Implemented a Docker-driven master-worker architecture for handling academic and competition submissions.
- Applied React and Material UI for front-end development; Django and PostgreSQL for back-end.
- Played a key role in establishing the submission system infrastructure for the ARCH project.

Blickfeld GmbH

Software Developer Intern · Full-time

October 2021 – March 2022

Munich, Germany

- Developed Django REST API endpoints with optimized algorithms and built React applications consuming them.
- Gained practical experience with Docker and Git CI pipelines.

PROJECTS

Personal Portfolio Website | *Next.js, Astro, Tailwind CSS, MDX* | Live · Source V1 · Source V2

2025 – Present

- Built across two iterations: V1 with Next.js 15, TypeScript, Convex real-time backend, Tiptap rich-text editing, and Framer Motion animations with a dynamic admin panel. V2 rebuilt with Astro, Tailwind CSS v4, and MDX content collections for a static, content-first approach.

FlexfitAI | *TUM DevOps Course Project* | Source

March 2025 – October 2025

- Key contributor to a team project designing and implementing a full-stack web application based on modern DevOps principles, featuring a Generative AI component.
- Engineered a Retrieval-Augmented Generation (RAG) system using Python, LangChain, and Weaviate for the GenAI module.
- Implemented containerization for all application microservices (client, server, GenAI, DB) using Docker and Docker Compose.
- Developed and managed CI/CD pipelines with GitHub Actions for automated building, testing, and deployment to a Kubernetes staging environment.
- Orchestrated deployment using Kubernetes (Helm) and configured observability with Prometheus and Grafana.
- Stack: React, Spring Boot, Python, Docker, Kubernetes, GitHub Actions, Prometheus, Grafana.

CommonRoad & ARCH Submission Systems | Project Link · Source

November 2022 – March 2024

- Contributed to projects using Django, React, S3 Buckets, PostgreSQL, RabbitMQ, Swagger, Docker, GitLab CI.
- Master-Worker Architecture systems for processing, grading, and storing submissions, and facilitated user competitions.

Connectz SetList Integration with Spotify | Project Link

October 2023 – October 2024

- Participated in a TUM Interdisciplinary Project aimed at developing an internal tool for the Connectz platform.
- Enabled users to create their own setlists using Spotify, leveraging Angular and Node.js in the technology stack.

E-Farmers

January 2021 – June 2021

- Developed an e-commerce graduation project, featuring a React.js frontend and Django REST API backend.
- Integrated smart box design using Arduino, with PostgreSQL and Amazon RDS for database management, Amazon S3 for storage, and Heroku for deployment.

SKILLS

Programming Languages: Python (primary), C/C++, JavaScript/TypeScript; familiar with Java, C#

Frameworks & Libraries: LangGraph, LangChain, Django, React, Angular, Spring Boot, Node.js

Tools & Cloud Technologies: Git, GitHub Actions, GitLab CI, Docker, Kubernetes, Helm, Prometheus, Grafana, SQL (PostgreSQL, MySQL, DuckDB, BigQuery), AWS (S3, RDS, Lambda), Terraform, Weaviate, AI-assisted development (GitHub Copilot, Claude Code, Codex)

Concepts & Methodologies: Agentic AI (Multi-Agent Systems, Tool Calling, MCP), Prompt Engineering, RAG, Text-to-SQL & Semantic Layers, LLM Evaluation (benchmark harnesses, labelled question sets), RESTful APIs, Microservices, DevOps (CI/CD), Observability, Agile/Scrum

Academic Interests: AI & Deep Learning, Large Language Models & Agents, Supply Chain AI, Resilient Cognitive Systems

Languages: Turkish (native), English (C1 certified), German (A2 in use)

Hobbies & Interests: Game Development (Godot 4), Web Applications, Linux Systems